# MultiQT notes

November 9, 2026

## 1 Methodological notes

### 1.1 Studentized score

Writing the score as a signed residual gives the usual additive MultiQT correction. Studentizing that residual gives multiplicatively scaled offsets, or, equivalently, offsets tracked in scaled coordinates. A changing scale rescales the accumulated correction as well as each new increment. If the scale decreases after a peak, it can shrink the outcome-scale contribution of a positive correction accumulated during an upswing. This is the motivation; the hidden accumulated state is not erased.

With quantile-specific scales, the ordering constraint defines a diagonally transformed, shifted isotonic cone. This is generally not a cone itself, but it remains a nonempty closed convex polyhedron with a unique Euclidean projection, computable through weighted isotonic regression in forecast coordinates. The gradient steps use a weighted quantile loss. The coverage criterion and calibration target are unchanged (strictly increasing invertible transformation); calibration is not guaranteed.

#### 1.1.1 Standard MultiQT

For standard MultiQT [1], let $\mathcal{A} = \{\alpha_1 < \cdots < \alpha_m\} \subset (0,1)$ and let $b_t = (b_t^\alpha)_{\alpha\in\mathcal{A}}$ be the base quantile forecasts available before observing $y_t$. Define

$$\mathcal{K} = \{q \in \mathbb{R}^m : q^{\alpha_1} \le \cdots \le q^{\alpha_m}\}, \qquad \Pi_C(x) = \arg\min_{z\in C} \|z - x\|^2,$$

where $C$ is a nonempty closed convex set. Assume $b_t \in \mathcal{K}$. The original MultiQT procedure is

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t^{\text{std}} = b_t + \theta_t, \tag{1}$$

$$\text{cov}_t^{\text{std},\alpha} = \mathbf{1}\{y_t \le q_t^{\text{std},\alpha}\}, \qquad \tilde{\theta}_{t+1}^\alpha = \tilde{\theta}_t^\alpha - \eta(\text{cov}_t^{\text{std},\alpha} - \alpha). \tag{2}$$

Here $\tilde{\theta}_t$ is the hidden offset and $\theta_t$ is the played (projected) offset. Both have outcome units.

#### 1.1.2 Score-threshold tracking

To make the connection explicit, first consider the scalar score-threshold update of Angelopoulos et al. [2]. We then recover the MultiQT update using the signed residual as the score.

A score function $S_t(y)$ maps a candidate outcome to a real number. Let $r_t = S_t(y_t)$ be the observed score and $c_t$ its issued threshold. The corresponding prediction set is

$$\mathcal{P}_t = \{y : S_t(y) \le c_t\}, \qquad \text{err}_t = \mathbf{1}\{r_t > c_t\}. \tag{3}$$

For target miscoverage $\delta$, update according to

$$\begin{aligned} c_{t+1} &= c_t + \eta(\text{err}_t - \delta) \\ &= c_t - \eta(\mathbf{1}\{r_t \le c_t\} - \alpha). \end{aligned} \tag{4}$$

The second line follows from $\mathrm{err}_t = 1 - \mathbf{1}\{r_t \le c_t\}$ and setting $\delta = 1 - \alpha$. The threshold $c_t$ is measured in score units and determines a prediction set through score inversion. For the signed affine scores below, that set's endpoint is the outcome quantile forecast.

Define the quantile loss by

$$\rho_\alpha(z, r) = \alpha(r - z)_+ + (1 - \alpha)(z - r)_+. \tag{5}$$

Holding the observed score $r$ fixed, its subdifferential in the threshold is

$$\partial_z \rho_\alpha(z, r) = \begin{cases} \{-\alpha\}, & z < r, \\ [-\alpha, 1 - \alpha], & z = r, \\ \{1 - \alpha\}, & z > r. \end{cases} \tag{6}$$

Selecting $1 - \alpha$ at equality gives

$$g_\alpha(z; r) = \mathbf{1}\{r \le z\} - \alpha \ \in \ \partial_z \rho_\alpha(z, r). \tag{7}$$

Thus $c_{t+1} = c_t - \eta g_\alpha(c_t; r_t)$ is (4). The score's distance from the threshold affects the loss; the side of the threshold determines the update direction and size.

### 1.1.3 MultiQT in residual coordinates

For each level, define the score and its observed value by

$$S_t^{\mathrm{std},\alpha}(y) = y - b_t^\alpha, \qquad r_t^{\mathrm{std},\alpha} = y_t - b_t^\alpha. \tag{8}$$

Using $\theta_t^\alpha$ as the score threshold gives

$$\{y : S_t^{\mathrm{std},\alpha}(y) \le \theta_t^\alpha\} = (-\infty, b_t^\alpha + \theta_t^\alpha] = (-\infty, q_t^{\mathrm{std},\alpha}]. \tag{9}$$

Consequently,

$$\mathrm{cov}_t^{\mathrm{std},\alpha} = \mathbf{1}\{y_t \le b_t^\alpha + \theta_t^\alpha\} = \mathbf{1}\{r_t^{\mathrm{std},\alpha} \le \theta_t^\alpha\}. \tag{10}$$

Substitution into (2) gives

$$\boxed{\tilde{\theta}_{t+1}^\alpha = \tilde{\theta}_t^\alpha - \eta(\mathbf{1}\{r_t^{\mathrm{std},\alpha} \le \theta_t^\alpha\} - \alpha).} \tag{11}$$

The original MultiQT loss has the exact residual-coordinate representation

$$\begin{aligned} \ell_t^{\mathrm{std}}(\theta) &= \sum_{\alpha \in \mathcal{A}} \rho_\alpha(b_t^\alpha + \theta^\alpha, y_t) \\ &= \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\theta^\alpha, y_t - b_t^\alpha) = \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\theta^\alpha, r_t^{\mathrm{std},\alpha}). \end{aligned} \tag{12}$$

The threshold $c_t$ of the scalar score tracker corresponds to the offset $\theta_t^\alpha$ when the score is $y - b_t^\alpha$. For a single level the constraint is all of $\mathbb{R}$, so the hidden and played states coincide and (11) is exactly scalar quantile tracking. For multiple levels, MultiQT uses a joint projection: evaluate $g_\alpha$ at the played $\theta_t^\alpha$, but apply the increment to the hidden $\tilde{\theta}_t^\alpha$.

### 1.1.4 Studentized residuals and their update

Now replace the residual score by the studentized residual.

Choose finite scales $\sigma_t^\alpha > 0$ from information available at issuance. They are fixed when the threshold is optimized. Define

$$S_t^{\mathrm{stu},\alpha}(y) = \frac{y - b_t^\alpha}{\sigma_t^\alpha}, \qquad r_t^{\mathrm{stu},\alpha} = \frac{y_t - b_t^\alpha}{\sigma_t^\alpha}. \tag{13}$$

Use $\tilde{u}_t$ and $u_t$ for hidden and played thresholds in studentized units. Inverting the score gives

$$S_t^{\mathrm{stu},\alpha}(y) \le u_t^\alpha \quad \Longleftrightarrow \quad y \le b_t^\alpha + \sigma_t^\alpha u_t^\alpha,$$
$$q_t^{\mathrm{stu},\alpha} = b_t^\alpha + \sigma_t^\alpha u_t^\alpha, \qquad \mathrm{cov}_t^{\mathrm{stu},\alpha} = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\}. \tag{14}$$

Standard MultiQT tracks the outcome-unit residual threshold $\theta_t^\alpha$ and issues $b_t^\alpha + \theta_t^\alpha$. Studentization tracks $u_t^\alpha$ and issues $b_t^\alpha + \sigma_t^\alpha u_t^\alpha$. The factor $\sigma_t^\alpha$ enters through the inverse score map.

Apply the same quantile-loss construction to the studentized scores:

$$\begin{aligned}\ell_t^{\mathrm{stu}}(u) &= \sum_{\alpha\in\mathcal{A}} \rho_\alpha(u^\alpha, r_t^{\mathrm{stu},\alpha}) \\ &= \sum_{\alpha\in\mathcal{A}} \frac{1}{\sigma_t^\alpha}\rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t).\end{aligned} \tag{15}$$

Using (7), its selected subgradient is

$$g_t^{\mathrm{stu},\alpha}(u_t) = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\} - \alpha = \mathrm{cov}_t^{\mathrm{stu},\alpha} - \alpha. \tag{16}$$

The chain rule gives the same result. Away from ties, for one level,

$$\begin{aligned}\frac{\partial}{\partial u^\alpha}\left[\frac{\rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t)}{\sigma_t^\alpha}\right] &= \frac{1}{\sigma_t^\alpha}(\mathbf{1}\{y_t \le b_t^\alpha + \sigma_t^\alpha u^\alpha\} - \alpha)\sigma_t^\alpha \\ &= \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u^\alpha\} - \alpha.\end{aligned} \tag{17}$$

At a tie use the selected subgradient in (7). The inverse scale in the loss cancels the scale in the inverse score map.

Retaining MultiQT's lazy update therefore gives

$$\boxed{\tilde{u}_{t+1}^\alpha = \tilde{u}_t^\alpha - \eta g_t^{\mathrm{stu},\alpha}(u_t) = \tilde{u}_t^\alpha - \eta\left(\mathbf{1}\left\{\frac{y_t - b_t^\alpha}{\sigma_t^\alpha} \le u_t^\alpha\right\} - \alpha\right).} \tag{18}$$

Initialize $\tilde{u}_1 = 0$. The score is compared with the *played* threshold; the increment is applied to the *hidden* threshold. The projection that produces $u_t$ is specified next.

| | Standard MultiQT | Studentized extension |
|---|---|---|
| Observed score | $r_t^{\mathrm{std},\alpha} = y_t - b_t^\alpha$ | $r_t^{\mathrm{stu},\alpha} = (y_t - b_t^\alpha)/\sigma_t^\alpha$ |
| Gradient | $\mathbf{1}\{r_t^{\mathrm{std},\alpha} \le \theta_t^\alpha\} - \alpha$ | $\mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\} - \alpha$ |
| Update origin | $\tilde{\theta}_t^\alpha$ | $\tilde{u}_t^\alpha$ |
| Issued forecast | $b_t^\alpha + \theta_t^\alpha$ | $b_t^\alpha + \sigma_t^\alpha u_t^\alpha$ |

The gradient has the same form because both losses apply quantile loss to a score threshold. The observed scores, state units, and issued forecasts change. An unnormalized outcome-unit loss would instead have gradient $\sigma_t^\alpha(\mathrm{cov}_t^{\mathrm{stu},\alpha} - \alpha)$ with respect to $u^\alpha$, and would define a different update.

### 1.1.5 Ordering constraints and projection

The constraint always concerns the outcome forecasts: $q_t^{\alpha_1} \le \cdots \le q_t^{\alpha_m}$. For the ordinary residual score, this requires

$$b_t + \theta \in \mathcal{K} \quad \Longleftrightarrow \quad \theta \in C_t^{\mathrm{std}} := \mathcal{K} - b_t, \tag{19}$$
$$\theta^{\alpha_i} - \theta^{\alpha_{i+1}} \le b_t^{\alpha_{i+1}} - b_t^{\alpha_i}, \qquad i = 1, \ldots, m-1. \tag{20}$$

For studentization, set

$$A_t = \mathrm{diag}(\sigma_t^{\alpha_1}, \ldots, \sigma_t^{\alpha_m}), \qquad q_t^{\mathrm{stu}} = b_t + A_t u_t.$$

Then

$$\boxed{C_t^{\mathrm{stu}} = \{u \in \mathbb{R}^m : b_t + A_t u \in \mathcal{K}\} = A_t^{-1}(\mathcal{K} - b_t).} \tag{21}$$

This is a closed convex polyhedron given by linear inequalities. It contains zero when $b_t \in \mathcal{K}$. The shift means that, like $\mathcal{K} - b_t$, it is generally not a cone. For unequal scales, $A_t^{-1}\mathcal{K} \neq \mathcal{K}$ in general, so the set cannot be simplified to $\mathcal{K} - A_t^{-1}b_t$. The standard constraint $b_t + \theta \in \mathcal{K}$ is replaced by $b_t + A_t u \in \mathcal{K}$, and the Euclidean projection is well defined:

$$u_t = \Pi_{C_t^{\mathrm{stu}}}(\tilde{u}_t), \qquad q_t^{\mathrm{stu}} = b_t + A_t u_t. \tag{22}$$

Substituting $u = A_t^{-1}(q - b_t)$ into the projection objective yields

$$\begin{aligned} q_t^{\mathrm{stu}} &= \arg\min_{q \in \mathcal{K}} \|A_t^{-1}(q - b_t) - \tilde{u}_t\|^2 \\ &= \arg\min_{q \in \mathcal{K}} \sum_{\alpha \in \mathcal{A}} \frac{(q^\alpha - b_t^\alpha - \sigma_t^\alpha \tilde{u}_t^\alpha)^2}{(\sigma_t^\alpha)^2}. \end{aligned} \tag{23}$$

Define $\Pi_{\mathcal{K}}^W(v) = \arg\min_{q \in \mathcal{K}} (q - v)^\top W (q - v)$ for a positive-definite matrix $W$. The exact identity is

$$\boxed{b_t + A_t \Pi_{A_t^{-1}(\mathcal{K} - b_t)}(\tilde{u}_t) = \Pi_{\mathcal{K}}^{A_t^{-2}}(b_t + A_t \tilde{u}_t).} \tag{24}$$

Thus the implementation is weighted isotonic regression of $v_t^\alpha = b_t^\alpha + \sigma_t^\alpha \tilde{u}_t^\alpha$ with weights $w_t^\alpha = 1/(\sigma_t^\alpha)^2$.

The ordinary residual case has forecast-space metric $I$; the studentized case has metric $A_t^{-2}$. These weights follow from the chosen Euclidean geometry of the tracked score coordinates.

### 1.1.6 Common and constant scales

For $S_t^\alpha(y) = (y - b_t^\alpha)/\sigma_t$, we have $A_t = \sigma_t I$. Positive scalar multiplication preserves $\mathcal{K}$:

$$C_t^{\mathrm{stu}} = \mathcal{K} - b_t/\sigma_t, \qquad q_t^{\mathrm{stu}} = \Pi_{\mathcal{K}}(b_t + \sigma_t \tilde{u}_t). \tag{25}$$

All projection weights are equal and cancel. Equivalently, run standard MultiQT on base $b_t/\sigma_t$ and outcome $y_t/\sigma_t$, then multiply the issued forecast by $\sigma_t$.

If $\sigma_t = \sigma$ is constant, standard MultiQT on the original inputs is also equivalent after converting the state and learning rate:

$$\tilde{\theta}_t = \sigma \tilde{u}_t, \qquad \eta_{\mathrm{std}} = \sigma \eta_{\mathrm{stu}}. \tag{26}$$

## 1.2 Questions and ideas

- What other scores would be reasonable to consider, i.e. what property of the base forecast do I need to recalibrate? Instead of asking for 1d PIT calibration, could I ask for 1d hdr-calibration? With studentized scores, or in general strictly increasing and invertible scores, the coverage indicator and calibration target do not change

# 2 Coding Updates

## 2.1 9 November 2026

### 2.1.1 Feedback handling: Stateful, re-run and delay

The feedback design is independent of the recalibration method. For a fixed disease, location and horizon, $s, t$ index forecast reference weeks, ordered by `reference_date` (Saturday); $q_s$ denotes the forecast for reference week $s$, with quantiles $q_s^\alpha$. Both report and forecast tables identify the target week by `target_end_date`. Reports additionally have a publication date, `version`: $y_s^{[v]}$ is the reported `value` for the target week of $q_s$ from the latest `version` on or before $v$. For forecast $q_t$, only reports available by the cutoff $\iota_t$ (`reference_date` minus three days, the Wednesday of the reference week, with few exceptions) influence $q_t$. A valid report is finite and nonnegative; $d \geq 0$ is extra delay in calendar weeks.

**Stateful (learn once).** Let $\nu_s$ be the `version` of the first valid report for the target week of $q_s$. Wait $d$ calendar weeks from that date. At the next forecast cutoff $\iota_t$ with $t > s$, look up the latest report. If it is invalid, wait for a later forecast; never substitute an older report.

$$m_s^{(d)} = \nu_s + 7d \text{ days}, \tag{27}$$

$$\tau_d(s) = \min\{t > s : \iota_t \geq m_s^{(d)},\ y_s^{[\iota_t]} \text{ is valid}\}, \qquad \mathcal{R}_t^{(d)} = \{s : \tau_d(s) = t\}. \tag{28}$$

Here $\tau_d(s)$ is the reference-week index $t$ when feedback for $q_s$ is used. The set $\mathcal{R}_t^{(d)}$ contains the earlier indices $s$ whose feedback is used before issuing $q_t$; an empty set means no update. For $s \in \mathcal{R}_t^{(d)}$, the update uses $y_s^{[\iota_t]}$, the latest report available by the cutoff of $q_t$. If a missing forecast postpones the update so that $\iota_t > m_s^{(d)}$, select the report at $\iota_t$, rather than fixing it at $m_s^{(d)}$. Feedback for each $q_s$ is used at most once; revisions after that update are ignored.

For plain MultiQT on counts, start at $\tilde{\theta}_0 = 0$; when learning is active, update using previously issued $q_s^\alpha$, then issue $q_t$:

$$\tilde{\theta}_t^\alpha = \tilde{\theta}_{t-1}^\alpha - \eta_t \sum_{s \in \mathcal{R}_t^{(d)}} (\mathbf{1}\{y_s^{[\iota_t]} \leq q_s^\alpha\} - \alpha), \tag{29}$$

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t = b_t + \theta_t. \tag{30}$$

Sum with $\eta_t$; keep the hidden state unprojected. By default, estimate $S_0$ in hindsight and run plain MultiQT from the start. Log tracking uses $\log(1 + y)$.

**Weekly re-run.** At each reference-week index $t$, re-run MultiQT from the start using the latest reports $y_s^{[\iota_t]}$. Only the new forecast $q_t^{[t]}$ is issued; previously issued forecasts remain unchanged. Earlier forecasts $q_s^{[t]}$ are reconstructed internally. Starting from $\tilde{\theta}_0^{[t]} = 0$, replay $j = 1, \ldots, t$ with feedback only at $j = \tau_d(s)$, retaining delay $d$; omit invalid reports:

$$\tilde{\theta}_j^{\alpha,[t]} = \tilde{\theta}_{j-1}^{\alpha,[t]} - \eta_j^{[t]} \sum_{\substack{s \in \mathcal{R}_j^{(d)} \\ y_s^{[\iota_t]} \text{ valid}}} (\mathbf{1}\{y_s^{[\iota_t]} \leq q_s^{\alpha,[t]}\} - \alpha), \tag{31}$$

$$\theta_j^{[t]} = \Pi_{\mathcal{K}-b_j}(\tilde{\theta}_j^{[t]}), \qquad q_j^{[t]} = b_j + \theta_j^{[t]}. \tag{32}$$

The stateful $S_0$ and expert loss-normalization scale, estimated from the first 20 released feedback items in hindsight, stay fixed across re-runs.

Studentized methods use $\tilde{u}$, $q^{\text{stu}}$ and projection (22), issuing the base until their scale is available; re-runs reconstruct the scale history. Evaluation uses a common latest-report truth, separate from learning.

#### 2.1.2 Effect of increasing feedback delay

#### 2.1.3 US and the five largest states

#### 2.1.4 Extended data history

#### 2.1.5 Trade-off and evaluation plots, including forecast curves

#### 2.1.6 Studentized scores and non-additive updates

#### 2.1.7 A small set of phase-specific experts

#### 2.1.8 Learned weights shared across quantile levels

## References


[1] Tiffany Ding, Isaac Gibbs, and Ryan J. Tibshirani. *Calibrated Multi-Level Quantile Forecasting*. arXiv:2512.23671v1, 2025. MultiQT paper.

[2] Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. *Conformal PID Control for Time Series Prediction*. Advances in Neural Information Processing Systems 36, 2023. NeurIPS paper.

[3] CMU Delphi. *Delphi hub forecasting: calibration experiments and tracker code*. Pinned source revision `9abebcfb`, inspected 9 November 2026. E09: constant learning rate and revision delay; tracker implementation.